DSS Application Design Document

Group 198

September 26, 2025

Contents

1 Overview

TBD.

2 Milestone Design Document (50%)

2.1 Application Overview

TBD.

2.2 Message Formats for Implemented Commands

This milestone uses a common message envelope for all requests and responses. Individual commands define their own bodies and semantics.

2.2.1 Envelope Format (Common to All Messages)

Envelope

```
{
  "version": 1,
  "message_id": "<role>-<pid>-<timestamp_ms>-<counter>",
  "message_type": "<command_or_response>",
  "status_code": "SUCCESS | FAILURE (optional on requests)",
  "reason": "Optional explanation on FAILURE",
  "body": { /* command-specific fields */ }
}
```

Message ID Format
Pattern: <role>-<pid>-<timestamp_ms>-<counter> (e.g., `user1-4123-1732665001123-00042`).

- **Role**: user name or disk name (e.g., `manager`, `user1`, `diskA`)
- **PID**: process ID of the sender
- **Timestamp**: current time in milliseconds
- **Counter**: incremented each time this process sends a message

Purpose: match requests and responses, enable retries on UDP, deduplicate requests, and simplify logging.

2.2.2 Command: `register_user`

Purpose

Register a user process with the manager.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user",
  "body": {
    "user_name": "User1",
```

```
    "ipv4_address": "192.0.2.10",
    "management_port": 2101,
    "command_port": 2102
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user_response",
  "status_code": "SUCCESS",
  "reason": null
}
```

On failure: `status_code: "FAILURE"` with a `reason`. Manager ensures unique user name and ports.

2.2.3 Command: `register_disk`

Purpose

Register a disk process with the manager.

Request

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk",
  "body": {
    "disk_name": "DiskA",
    "ipv4_address": "192.0.2.11",
    "management_port": 2201,
    "command_port": 2202
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk_response",
  "status_code": "SUCCESS"
}
```

Manager marks the disk as `Free`.

2.2.4 Command: `configure_dss`

Purpose

Configure a DSS from available disks.

Request


```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss",
  "body": {
    "dss_name": "DSS1",
    "number_of_disks": 3,
    "striping_unit_bytes": 1024
  }
}
```


Response (success)


```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss_response",
  "status_code": "SUCCESS",
  "body": {
    "dss_name": "DSS1",
    "disk_order": ["DiskA", "DiskB", "DiskC"],
    "striping_unit_bytes": 1024
  }
}
```


On failure: `status_code: "FAILURE"` with a `reason`. Manager validates all rules.

2.2.5 Command: `deregister_user`

Purpose

Deregister a user process from the manager.

Request


```
{
  "version": 1,
  "message_id": "user1-4123-1732665002500-00003",
  "message_type": "deregister_user",
  "body": {
    "user_name": "User1"
  }
}
```

Response


```
{
  "version": 1,
  "message_id": "user1-4123-1732665002500-00003",
  "message_type": "deregister_user_response",
  "status_code": "SUCCESS"
}
```


2.2.6 Command: deregister_disk

Purpose

Deregister a disk process from the manager.

Request


```
{
  "version": 1,
  "message_id": "diskA-6123-1732665002700-00004",
  "message_type": "deregister_disk",
  "body": {
    "disk_name": "DiskA"
  }
}
```


Response


```
{
  "version": 1,
  "message_id": "diskA-6123-1732665002700-00004",
  "message_type": "deregister_disk_response",
  "status_code": "SUCCESS"
}
```


2.2.7 Manager Protocol Commands (TBD for Full Project)

Command	Status
ls	TBD
copy	TBD (Phase 1 and Phase 2)
read	TBD (Phase 1 and Phase 2)
disk_failure	TBD (Phase 1 and Phase 2)
decommission_dss	TBD

2.3 Time–Space Diagrams

2.3.1 Command: `register_user`

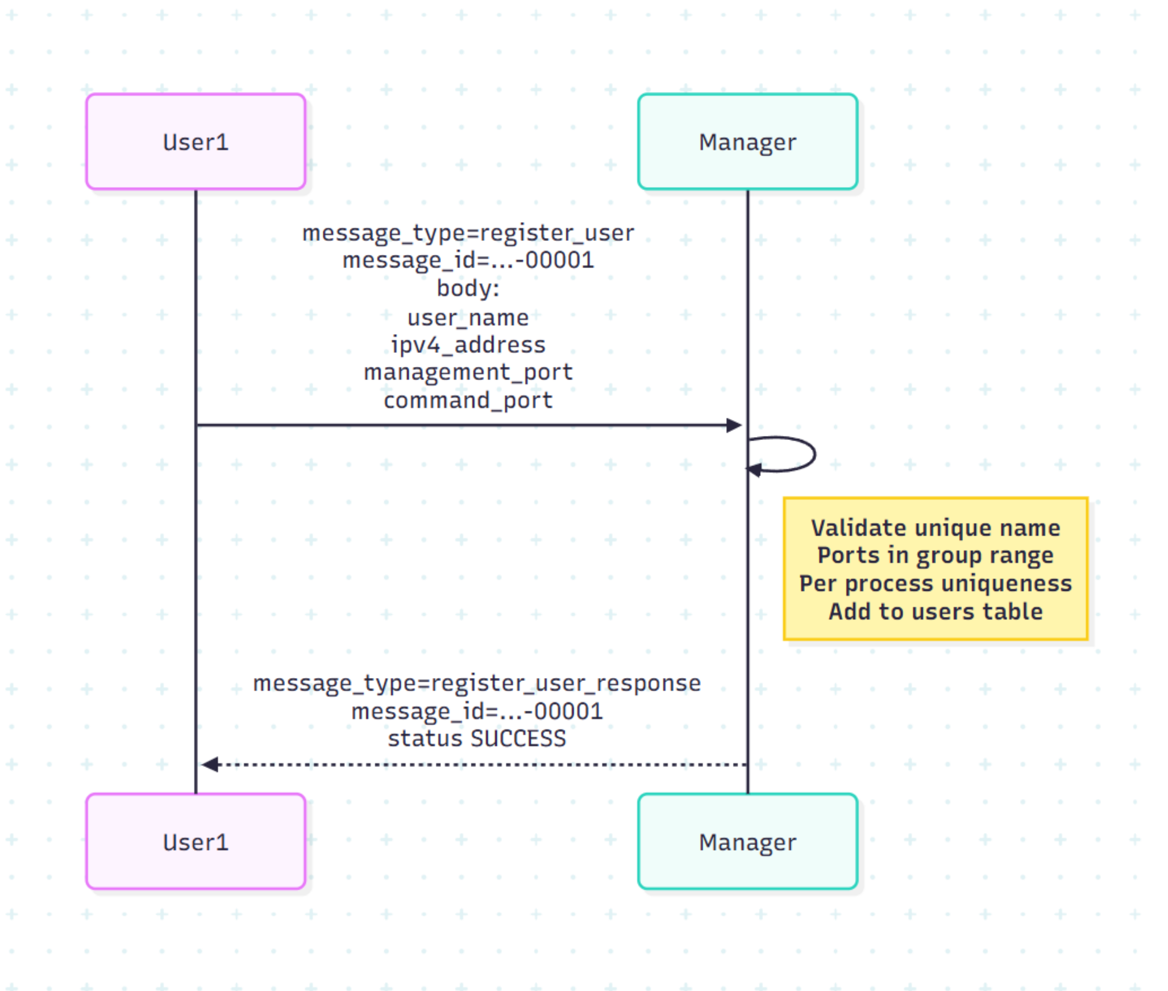

Figure 1: Time–space diagram for `register_user`

2.3.2 Command: `register_disk`

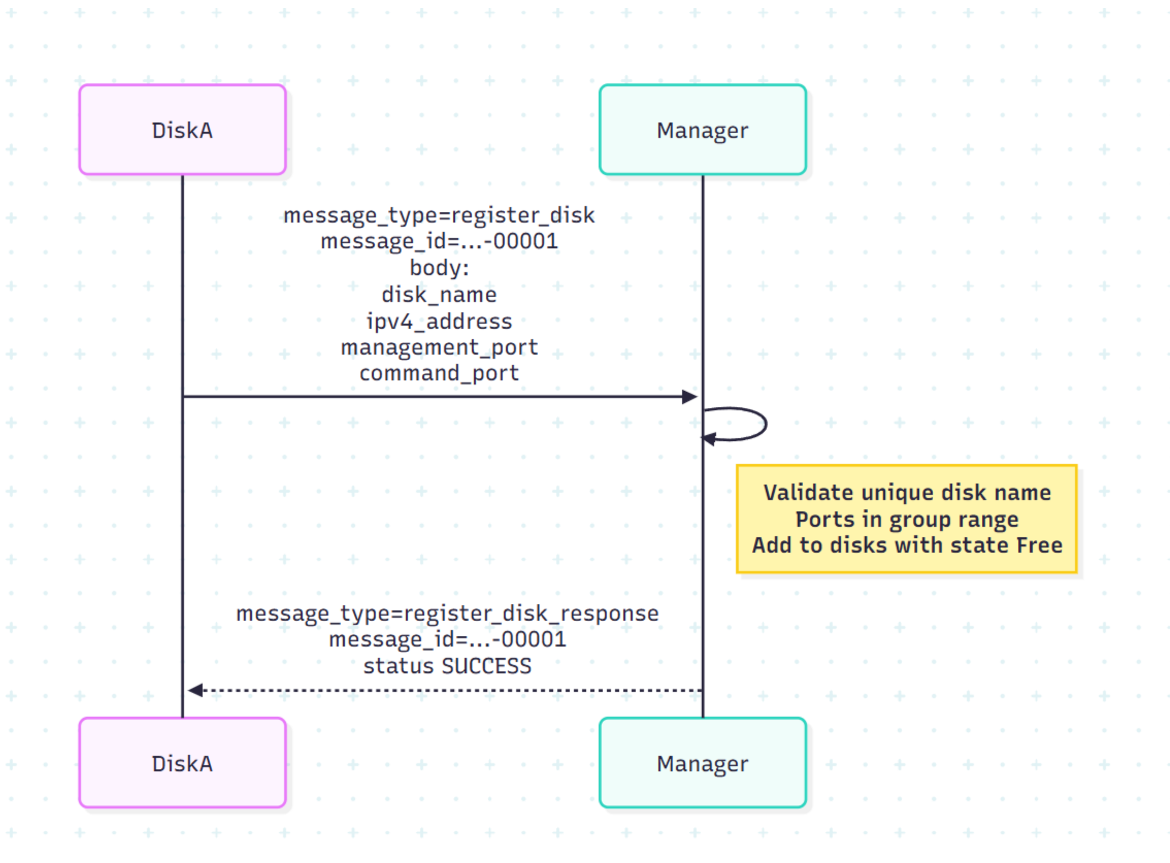

Figure 2: Time–space diagram for `register_disk`

2.3.3 Command: configure_dss

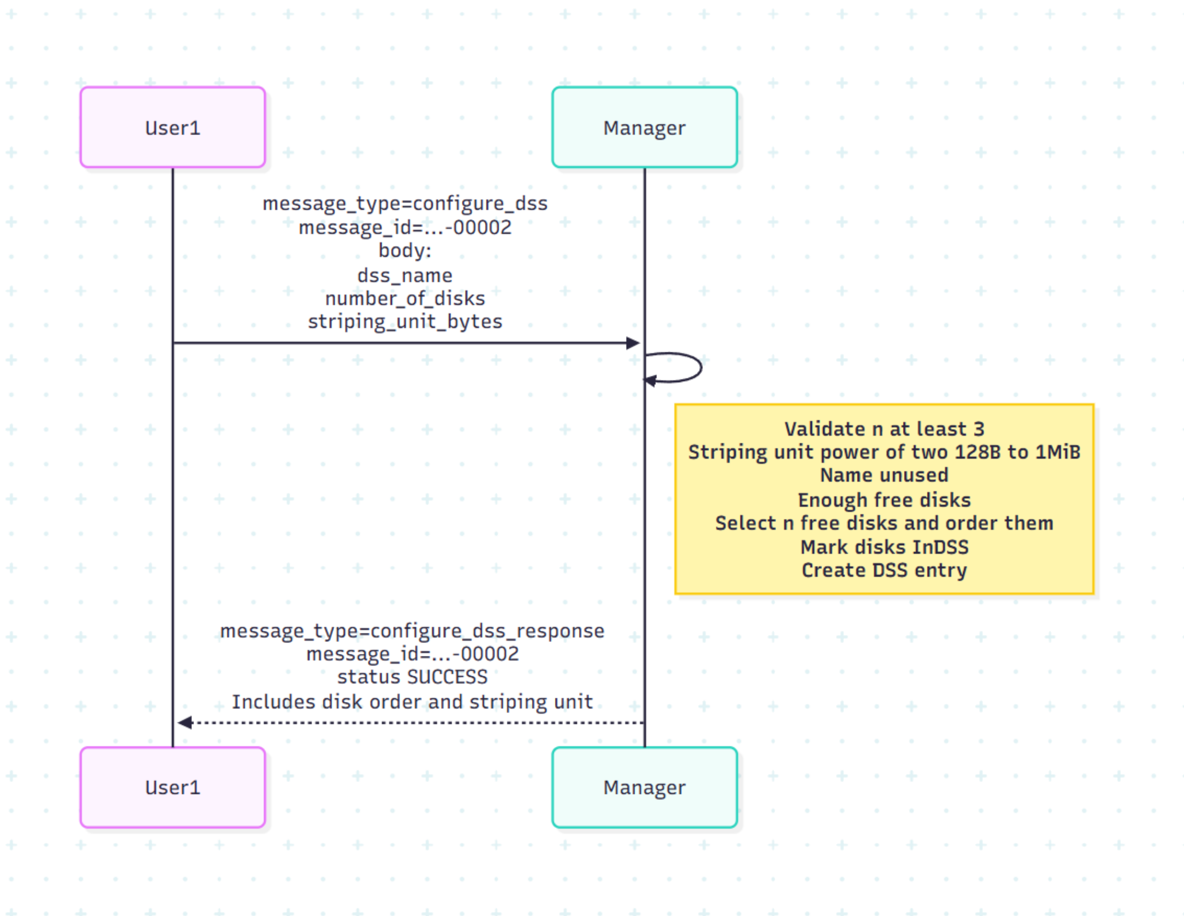

Figure 3: Time–space diagram for configure_dss

2.3.4 Command: `deregister_user`

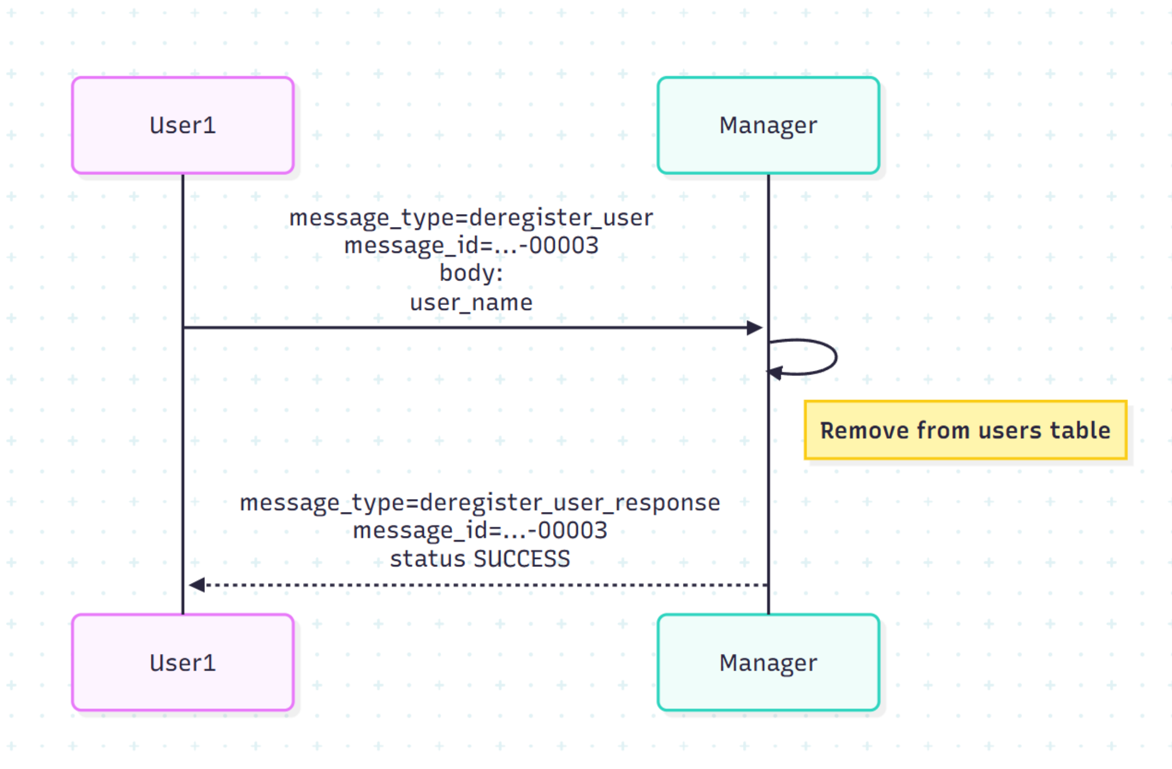

Figure 4: Time–space diagram for `deregister_user`

2.3.5 Command: `deregister_disk`

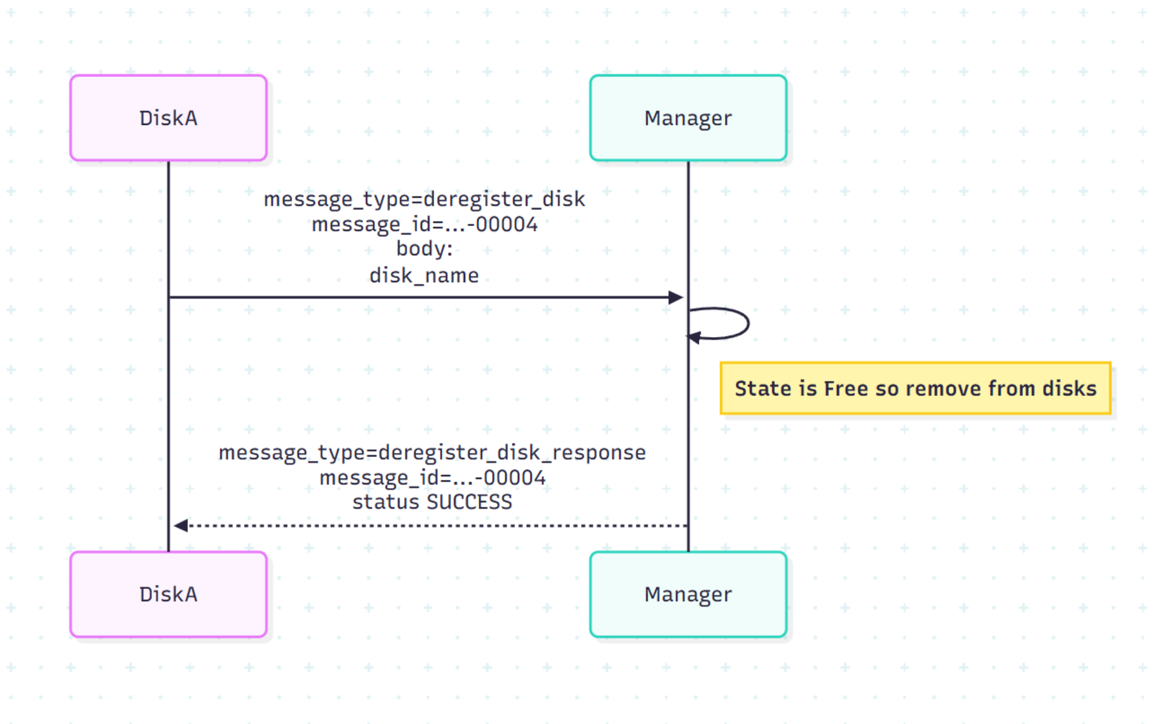

Figure 5: Time–space diagram for `deregister_disk`

2.4 Data Structures and Design Decisions

2.4.1 Manager State

TBD.

2.4.2 Client/User State

TBD.

2.5 Port Numbers

Because I am Group 198, the port numbers are 21200–21299 (inclusive).

2.6 Screenshots

Provide the following screenshots (place files under `screenshots/`):

i. Output of: `git log --pretty=format:"%h - %an, %ad (Commit) - %cd (Author)"` (TBD)
ii. Entire commit history of the GitHub repository (TBD)
iii. Output of: `git reflog` (TBD)
iv. Output of: `git fsck` (TBD)

TBD: Insert screenshot of git log command output

Figure 6: `git log` output

TBD: Insert screenshot of full commit history

Figure 7: Repository commit history

2.7 Video Demo Link and Timestamps (3(a)–3(f))

Accessible Link `TBD`

Timestamps TBD.
1. 3(a): TBD
2. 3(b): TBD
3. 3(c): TBD
4. 3(d): TBD
5. 3(e): TBD
6. 3(f): TBD

3 Extended Design for Full Project (30%)

3.1 Additional Commands Implemented

TBD.

3.1.1 Command: `<NAME>`

Purpose TBD.

Request Message Format TBD.

Response Message Format TBD.

Error Handling TBD.

3.2 Updated Design Decisions and Use of Threads

TBD.

3.3 Screenshots (Repeat Items from Step 1 of §4.1)

TBD.

3.4 Video Timestamps (3(a)–3(h))

1. 3(a): TBD
2. 3(b): TBD
3. 3(c): TBD

TBD: Insert screenshot of `git reflog` output

Figure 8: `git reflog` output

TBD: Insert screenshot of `git fsck` output

Figure 9: `git fsck` output

4. 3(d): TBD
5. 3(e): TBD
6. 3(f): TBD
7. 3(g): TBD
8. 3(h): TBD

A Message Format Template

TBD.

B Time–Space Diagram Notes

TBD.